

Ananta Nayak

AI / ML Engineer / Generative AI
Assam, India • +91-8822891692 • anantanayak337@gmail.com
github.com/Ananta01Nayk • linkedin.com/in/ananta-nayak

Summary

AI/ML Engineer with hands-on experience in Generative AI, Retrieval-Augmented Generation (RAG), data analytics, and legacy system modernization. Experienced in building end-to-end ML systems, agentic workflows, and AI-powered developer tools.

Skills

- **Programming Languages:** Python, JavaScript, SQL
- **Frameworks & Libraries:** TensorFlow, Keras, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib
- **Deep Learning Architectures:** CNN, RNN, LSTM, GRU, GAN, Transformer
- **NLP & Generative AI:** GPT APIs, LLMs, LangChain, LangGraph, Embeddings, Retrieval-Augmented Generation (RAG), NLTK
- **Machine Learning Concepts:** Supervised Learning, Unsupervised Learning, Feature Engineering, Model Evaluation, Deployment
- **Backend & APIs:** Flask, FastAPI
- **Databases:** MySQL, MongoDB, Vector Databases
- **Web & UI:** React.js, Streamlit, HTML, CSS, JavaScript
- **Developer & Platform Tools:** VS Code Extension APIs, Git, GitHub, GitLab
- **Data Analytics & BI:** Exploratory Data Analysis (EDA), Statistical Analysis, Power BI, Tableau, Excel

Experience

AI Engineering Intern — (Remote)

19 Jan 2026 – 13 Feb 2026

Letter of Appointment (LOA) • Internship Certificate

- Developing a **Generative AI-based ERPNext modernization system** for large legacy Python codebases.
- Building an **AI-powered VS Code extension** for contextual code understanding and refactoring insights.
- Implementing **RAG pipelines** using embeddings and vector databases.
- Designing scalable workflows to analyze repositories and generate structured technical outputs.

Data Science & Analytics Intern — (Remote)

Jun 2025 – Sep 2025

Internship Certificate

- Built end-to-end ML pipelines including data preprocessing, model training, and evaluation.
- Developed CNN, LSTM models for classification tasks.
- Performed in-depth **EDA and statistical analysis** to identify trends and data anomalies.
- Applied **feature engineering and data transformation techniques** to improve model performance.
- Created interactive **Power BI dashboards** to communicate insights to stakeholders.
- Optimized datasets using **outlier detection and missing value handling** techniques.
- Assisted in **model performance comparison and validation** using cross-validation metrics.

Projects

OmniScript Agent — Agentic RAG System

GitHub

Python, LangGraph, MongoDB, FastAPI

- Built an **agentic RAG system** for conversational interaction with complex PDF documents.
- Implemented **Self-RAG logic** to validate retrieval relevance and self-correct LLM outputs.
- Integrated **MongoDB Atlas Vector Search** for high-performance semantic retrieval.

AI Code Helper — Generative AI VS Code Extension

GitHub

Python, JavaScript, LLMs

- Developed an AI assistant to **understand entire codebases** inside VS Code.
- Detected **syntax errors, bugs, and business logic issues** using LLM reasoning.
- Generated file-level and project-level explanations for faster debugging.

Cancer Detection System Using CNN

GitHub

Python, TensorFlow, Flask, HTML, CSS, JavaScript

- Designed and trained a **CNN-based model** for cancer detection from medical images.
- Built a **Flask REST API** to serve real-time predictions.
- Developed a responsive web interface and achieved **92% accuracy**.

Education

MCA — Artificial Intelligence & Machine Learning (2023-2025)

Chandigarh University, Punjab

BCA (2017-2021) — Guwahati University, Assam

Certifications

Data Science & Analytics — Zidio Development • Data Analytics — Coursera